Failure-Driven Acquisition and Selection for Autonomous Simulation Optimization

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Abstract

Improving an embodied policy often requires acquiring the right data, not merely changing its architecture or optimizer. We develop an autonomous research workflow that connects development-set policy failures to executable data acquisition and selection hypotheses in simulation repositories. A research agent inspects native expert and conversion interfaces, proposes a data intervention, and executes it under resource and evaluation constraints. An evidence layer connects immutable data versions to actual training consumption, loaded policy identity, and native rollout measurements. We separate equal-accepted-data comparisons from equal-total-compute comparisons, and reserve independent evaluation conditions before optimization. This manuscript specifies the implemented evidence mechanisms and the experiments needed to test failure-directed acquisition against random expansion and fixed-data continuation. ***Development draft: real comparative experiments are pending; no performance improvement or SOTA claim is made.***

1. Introduction

Robot imitation learning depends on the distribution of demonstrations available to the learner. A policy can achieve low training loss and still fail at states underrepresented in its demonstrations. Simulation offers an opportunity to generate additional supervision, but a runnable simulator alone is not an expert: a usable acquisition pipeline also needs task initialization, valid expert actions, success and termination semantics, serialization, and a compatible training loader. Acquiring more trajectories therefore involves both a learning decision and a repository integration problem.

Scalable generation systems such as MimicGen provide powerful mechanisms for expanding demonstrations [3]. DataMIL demonstrates that selection should be evaluated through downstream policy performance rather than assumed useful from human notions of similarity [1]. RESample directly addresses missing recovery supervision through coverage-guided augmentation [4]. These works make a weak novelty claim—"failure data are useful"—insufficient. Our intended question is narrower: can an autonomous repository-level workflow turn observed development failures into valid, resource-accounted interventions that outperform equally resourced alternatives without human experiment steering?

The workflow builds on the research-agent decomposition of AutoSOTA [2], rather than claiming a new agent architecture. Repository acquisition replaces paper-to-code discovery. A specialized Data role investigates expert availability, acquisition conditions, quality checks, conversion, provenance, and training consumption. The main Scheduler remains responsible for decisions and execution; independent supervision checks scientific validity. Human-readable run reports expose what was actually attempted, what remains blocked, and what the evidence supports.

The intended contribution is an experimentally validated data intervention workflow, supported by: (1) executable, source-grounded acquisition and selection hypotheses; (2) an auditable chain from native collection to policy evaluation; and (3) controls distinguishing failure targeting, filtering, additional data, and additional compute. The first two currently have implementation and contract coverage. Whether the third yields a publishable empirical contribution remains an experimental question, not a consequence of passing software tests.

2. Related Work

Demonstration generation. MimicGen scales robot-learning data through generation from human demonstrations across environments and tasks [3]. We do not introduce a replacement motion planner or general expert. We investigate whether a research agent can identify and connect a repository's existing generation mechanism, then choose useful conditions within its supported training domain. A repository without a valid automated expert is out-

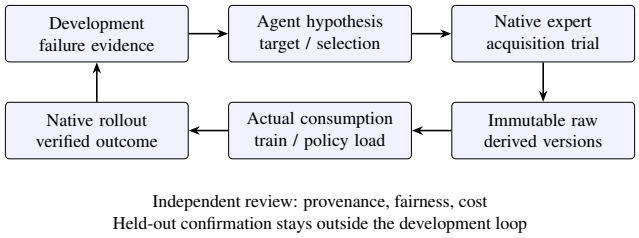


Figure 1. Evidence-grounded development loop. Each edge needs native evidence; an agent plan is not a successful acquisition or a verified performance gain.

side the autonomous acquisition track unless another legitimate supervision mechanism is explicitly available.

Performance-aware selection. DataMIL uses datamodel-based reasoning and a task-specific surrogate to select data for imitation learning [1]. Our present implementation does not include its datamodel algorithm. It is a relevant comparator when its requirements and cost fit the selected policy and task. We must establish whether failure-directed acquisition adds value beyond quality filtering and performance-aware selection of already available data.

Recovery supervision. RESample uses a conservative coverage function, exploratory sampling, and recovery actions to supplement missing corrective supervision [4]. Thus, failure-guided recovery collection itself is not an unclaimed direction. Our workflow must specify which native acquisition decisions are automated, what evidence selects them, and whether automation generalizes to previously unseen tasks or repositories. Unsupported actions are not relabeled as expert supervision.

3. Method and Implemented Evidence Layer

3.1. Problem and capability boundary

An input consists of an isolated source checkout, explicitly connected resources, a native development evaluation protocol, and a total resource envelope. Let D_0 denote available training demonstrations and π_0 the reproducible baseline. Development rollouts provide evidence E_t. The agent proposes an intervention $h_t = (g_t, q_t, m_t)$, where g_t describes supported acquisition conditions, q_t specifies selection, and m_t defines the training mixture. These are research decisions, not a hard-coded benchmark recipe. Their source connections must be established before execution.

The intended iteration is

$$D_t^+ = q_t(\mathrm{Collect}_{\mathrm{native}}(g_t)), \quad \pi_{t+1} = \mathrm{Train}(D_0, D_t^+; m_t). \tag{1}$$

The equation specifies an experiment, not a guarantee of a usable collector. A small native acquisition trial must establish expert validity, complete-episode output, serialization, and conversion before scaling. If these prerequisites cannot be met within resources, the run reports that boundary. It does not claim acquisition from a planned command or reinterpret failed learner actions as successful expert trajectories.

3.2. Agent decisions versus execution constraints

The Scheduler maintains a persistent research plan and consults Data, Resource, Init, Fix, and other roles as needed. Agents first see a short skill directory and explicitly choose relevant procedures with reasons. Recommendations do not silently replace agent choices. Data analyzes the connection between observed failure, native expert capabilities, target conditions, quantity, quality, and loader consumption. Resource reasons about local duration, storage, and cost; the Scheduler can submit long jobs, inspect progress, revise local limits, or cancel stalled jobs. The executor enforces only authorized total ceilings and experiment contracts.

This separation matters for portability: changing repositories should change source-derived commands and proposals, not require a new hidden benchmark branch. However, an unconstrained shell or self-modifying evaluator would invalidate results. Skills and agents cannot override held-out separation, metric semantics, unknown usage reservations, or immutable experiment evidence.

3.3. Data identity and actual consumption

Raw outputs are immutable research resources; filtering and conversion create derived versions with parent identities and transformation provenance. A recorded acquisition observation must have a verified native collection ancestor. Merely processing an existing official dataset does not qualify as new acquisition. The version registry connects each experimental arm to its acquired and training versions, while the measurement records actual training consumption.

The chain to be verified is

$$\begin{aligned}&\text{Native producer} \rightarrow \text{raw data} \rightarrow \text{derived view}\\ &\rightarrow \text{consumed training input} \rightarrow \text{loaded policy}\\ &\rightarrow \text{native rollout} \rightarrow \text{measured outcome}.\end{aligned}$$

Registering an intention or observing a checkpoint file is insufficient. Recorded observations require verified policy consumption, rollout evidence, metric lineage, and data consumption. One measurement cannot represent both targeted and random arms, or multiple seeds. Evidence is checked again during export; changed data or inconsistent native receipts exclude a row instead of producing a stale table.

3.4. Campaigns and independent review

A campaign records a development measurement, a falsifiable hypothesis, acquisition and selection plans, required arms, accepted-episode budget, declared training seeds, recipe, and cost-accounting intent. Its registration is not an experiment result. A campaign observation joins a new scored measurement to its native-produced data lineage. Scientific fairness is reviewed separately from whether an individual score is genuine. Missing or insufficient review remains uncertain; a real metric does not establish matched training exposure or fair cost.

3.5. Evaluation separation and failure recovery

Independent evaluation conditions are reserved before optimization, using source-supported native inputs. A training RNG seed alone does not establish disjoint initial conditions. Once search or measurement begins, the reservation cannot be retroactively replaced. Held-out outcomes are not used to propose acquisition targets or filter training data.

Execution failures are sealed with stable evidence identifiers. Fix receives the actual command, outcome, and evidence rather than a shortened display message. Valid replacements require a source/evidence connection and reverification of the failed operation. Framework development interventions are recorded separately; trajectories that required such interventions cannot be described as uninterrupted, frozen-system autonomous trials.

4. Experimental Protocol

4.1. Feasibility before scale

The first gate is a real native collection–conversion–training–evaluation cycle on a repository with an automated expert. The pilot uses source isolation and explicit read-only resource links, rather than copying every resource or assuming that copying code preserves its runtime. It must produce genuinely new usable episodes and show that the intended version reaches training. A runnable environment or an old baseline score does not pass this gate.

After the pilot, freeze the harness, skill corpus, model configuration, resource bindings, and protocol. Select additional tasks and an unseen repository according to verified expert capabilities and available resources. Record exclusions before viewing held-out results. Development repairs are allowed during feasibility but must not silently enter formal trials.

4.2. Controls and two resource tracks

Table 1 defines the minimum controls. All arms share the native policy family, initialization policy, official-data access, permitted training domain, metric semantics, and common quality floor. The targeted arms may differ in acquisition conditions, not in privileged test access or expert quality.

Arm	Purpose
Official recipe	Reproducible fixed-data reference.
Continuation	No-new-data, equal-total-compute control.
Random expansion	New data without failure targeting.
Targeted expansion	Targeting with the common quality floor.
Targeted + selection	Full intervention with task-directed selection.

Table 1. Preregistered controls. These are planned experiments, not completed results.

In the *equal-data* track, targeted and random arms receive the same number of accepted complete episodes. Report attempts, rejection rates, duplicates, conversion losses, actual training exposure, and acquisition cost. Equal accepted counts do not imply equal compute. In the *equal-compute* track, charge failed acquisition, conversion, training, evaluation, and research overhead against matched envelopes; fixed-data continuation receives the corresponding opportunity. GPU device-lease time, wall time, and model cost remain distinct quantities.

4.3. Replication and statistics

Use at least three independently trained seeds for the small confirmed task set, with matched initialization procedures. Training-seed replication with a fixed acquired dataset is different from independent replications of the complete agent acquisition loop. Report the former as policy variation, and evaluate the latter separately where affordable. Pooling many evaluation episodes is not a substitute for independent training seeds or independent agent trials.

Episode-level paired comparisons require recorded matching initial-state identities; shared episode indexes or RNG labels are insufficient. Otherwise use unpaired comparisons and state that limitation. Report effect sizes and uncertainty, not just the best run. A descriptive mean across declared seed IDs does not establish that those seeds were actually used; independent source and receipt audit is required.

4.4. Reporting and attribution

Each run maintains a Chinese human-readable report with status, actual scored comparisons, data progress, plots, and available native demos. Missing measurements are displayed as pending, not zero success. Paper exports are read-only, content- addressed snapshots. They distinguish whole-run costs from per-arm costs and development evidence from frozen formal evidence. Manual fixes, restarts,

research choices, and unknown usage reservations are disclosed.

5. Results and Analysis: Pending Real Experiments

The current implementation provides campaign provenance, consumption checks, confirmation reservation, independent review interfaces, and development exports. Software contract tests verify these mechanisms but are not embodied performance results. The active pilot has not yet supplied the comparative measurements needed for a results table. We therefore omit numeric claims rather than reuse historical debugging scores or fill missing experiments with zero.

The eventual analysis must answer: does targeting improve success at equal data quantity; does it retain an advantage at equal total cost; which failures are recoverable by the native expert; and do decisions transfer to held-out tasks or repositories without operator intervention? Ablations should remove failure targeting and task-directed selection separately, holding the remaining procedure constant. Additional policies or more repositories are useful only if the core comparison is already reproducible.

6. Limitations and Submission Gate

Automated acquisition is not supported by every simulation repository. Missing assets, incompatible hardware, absent expert supervision, or non-reproducible baselines may prevent experiments. Source citations do not prove successful runtime integration. Likewise, agent-generated plans do not prove that acquired data cover the identified failures or improve a learned policy.

The current pilot is a development run with documented framework intervention, not formal evidence of complete autonomy. A submission needs real equal-data comparisons, cost controls, independent replication, held-out evaluation, and a clear incremental mechanism beyond existing data selection and recovery methods. If those gates fail, the scope or submission timeline should change. The paper must not substitute a broad framework narrative for missing experimental evidence.

References

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